DATA1046B - (GROUP) ASSIGNMENT I

INSTRUCTIONS

The objective of this assignment is to design an ER diagram using Crow's Foot Notation to represent a hospital management system and implement the corresponding database using Data Definition Language statements in MySQL workbench.

Part 1: ER Diagram in MySQL workbench (*10 points*)

Your design must be based on the following business rules:

- A doctor can treat multiple patients.
- Each patient must be always assigned to one and only one doctor.
- A patient can make multiple appointments
- An appointment is made by exactly one patient.
- A doctor can have multiple appointments
- Each appointment must be associated with exactly one doctor.
- A doctor has exactly one specialization.
- Each specialization can be assigned to multiple doctors.
- A patient can receive multiple treatments.
- Each treatment can be provided to multiple patients.
- A doctor can administer multiple treatments
- A treatment can be administered by multiple doctors.

Your objectives:

1. Insert a table for each of the identified entities
2. Identify at least three potential **non-key attributes** per entity.
3. Identify potential relationships between each pair of entities (i.e. one-to-one, one-to-many, and many-to-many relationships).
4. Where applicable, decompose any many-to-many relationships into two one-to-many relationships to eliminate redundancy.
5. Make sure to use Crow's Foot Notation to create the ER diagram in MySQL workbench

Part 2: Database Implementation (*10 points*)

In the second part of the assignment, you will manually implement the database for the hospital management system using Data Definition Language statements in MySQL Workbench. **You are not allowed to use forward engineering or any AI-based tool to generate SQL code**.

Your objectives:

1. Create tables for each entity defined in the ER diagram
2. Define primary keys for each table

3. Establish foreign key relationships between tables, as indicated by the relationships in your ER diagram.
4. Use appropriate data types for each column and apply appropriate constraints.
5. Create bridge tables, when applicable.

Deliverables:

- A SQL file containing all the DDL statements that define that database schema.
- A mwb file containing the ER diagram

Teams:

- Groups of two students

Due Date:

- October 5, 2026 11:59 PM